

3-Mark Questions & Answers — DBMS Module 4: Relational Database Design

Q1. Compare 3NF and BCNF with a suitable example.

Property	3NF	BCNF
FD Condition $X \rightarrow Y$	X is Super Key OR Y is Prime Attribute.	X MUST be a Super Key (No exceptions).
Overlapping Keys	Allows overlapping candidate key dependency.	Eliminates overlapping key redundancy.
Dependency Preservation	Always Guaranteed.	Not always guaranteed.

- Example: `ADVISOR(Student_ID, Subject, Advisor_Name)` where `Advisor_Name → Subject`. It is in 3NF (Subject is prime) but NOT in BCNF (Advisor_Name is not a superkey).

Q2. Explain the algorithm for attribute closure X^+ .

Input: Set of attributes X, Set of FDs F
Output: X^+ (Attribute Closure)

1. Set $X^+ = X$
2. Repeat until X^+ does not change:
For each FD $(Y \rightarrow Z)$ in F:
If Y is a subset of X^+ :
 $X^+ = X^+ \cup Z$
3. Return X^+

Q3. Differentiate between functional dependency and multivalued dependency.

Feature	Functional Dependency (FD)	Multivalued Dependency (MVD)
Symbol	$X \rightarrow Y$	$X \twoheadrightarrow Y$
Mapping	Single value: X determines a unique single value of Y.	Multiple values: X independent multidetermines a set of Y values.
Normal Form Target	2NF, 3NF, BCNF	4NF

Q4. What are NoSQL data models? List its 4 categories.

NoSQL databases are non-relational, schema-less, horizontally scalable systems built for big data.

1. **Key-Value Stores:** Redis, Amazon DynamoDB
2. **Document Stores:** MongoDB, CouchDB

3. **Column-Family Stores:** Apache Cassandra, HBase

4. **Graph Databases:** Neo4j, Amazon Neptune

Q5. Explain dependency preservation in database decomposition.

A decomposition of R into sub-relations **R₁, R₂, ..., R_n** is **dependency preserving** if the closure of the union of all functional dependencies enforced locally on each sub-relation equals the closure of the original set of functional dependencies:

$$(F_1 \cup F_2 \cup \dots \cup F_n)^+ = F^+$$

It ensures database integrity constraints can be enforced without expensive JOIN operations across tables.
